

Assistive Communication Technologies and Stigma: How Perceived Visibility of Cochlear Implants Affects Self-Stigma and Social Interaction Anxiety

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Assistive technologies promise improved functionality for people with disabilities, but their use and adoption are complicated. Previous research has identified several reasons why people with disabilities choose to abandon their assistive technologies. However, less is known about the adoption of assistive technologies that are permanent and user perceptions of these technologies. This study builds upon existing assistive technology, disability, identity, and stigma literature by examining the role of perceptions of cochlear implants, an assistive technology that permanently renders an invisible disability visible, among the single-sided deaf community. Analysis of survey data from 332 single-sided deaf individuals with cochlear implants all over the world revealed that the perceived visibility of assistive technology is positively associated with self-stigma and social interaction anxiety. Results from this study highlight the centrality of sociotechnical perceptions of assistive technologies to the experience of stigma, rather than the objective characteristics and features of technology, and problematize the existing dichotomous approach (visible v. invisible) to understanding assistive technologies. This study also illuminates the importance of studying assistive technologies on an intragroup level as well as focusing on how specific stigmatized niche communities may perceive and use their assistive technologies.

CCS Concepts: • **Human-centered computing** → **Accessibility**, **Empirical studies in accessibility**

Additional Key Words and Phrases: Assistive technologies; cochlear implants; single-sided deafness; stigma; social interaction anxiety

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1 INTRODUCTION

Within the disability literature, a great deal of emphasis is placed on assistive technologies (AT). Defined as “any product, instrument, equipment or technology adapted or specially designed for improving the functioning of a disabled person” [101], AT can increase independence, improve accessibility, and ease social interactions for people with disabilities. However, individuals with various disabilities have expressed apprehension using AT and displaying their disability for fear they may be further stigmatized in public [13, 57, 103], experience disruption in their social relationships [39], and/or because they do not want to self-identify as disabled [22]. Hence, although

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AT can provide many functional benefits, there are several sociocultural and psychosocial concerns individuals have that may create barriers to accessing these positive outcomes [26, 53, 57, 66, 72, 80].

AT encompass an array of devices, ranging from temporary to permanent, voluntary to involuntary, and visible to invisible. Much of the existing literature on AT focuses on temporary, voluntary, and visible forms of AT (e.g., hearing aids, crutches, walking canes, glasses for the visually impaired, etc.) and how to improve adoption and combat non-use [29, 32, 33, 78, 93]. Many studies also explore motivations for intermittent use or abandonment of AT and whether they are physical, aesthetic, or the result of stigma associated with AT and AT users [27, 46, 49, 50, 91]. Based on the findings of these studies, several design solutions are being proffered, such as giving those with disabilities greater control over the aesthetics of AT [82], creating matching frameworks that healthcare providers can use to assist their patients with selecting and adopting AT [89], increasing participatory and user-centered design of AT [20, 61], and developing models to capture individuals' intention of using new technologies [77]. While these studies are important, they tend to focus on overcoming barriers to adoption and assume that increasing AT adoption and use among individuals with disabilities is not only beneficial, but ideal. Less work has examined permanent forms of AT, such as cochlear implants (CIs), which are surgically implanted and thus, cannot be easily removed or abandoned. Whereas individuals with temporary AT are able to make ongoing decisions about AT use, those with permanent AT are incapable of forgoing AT use, and thus, must continually contend with and manage the consequences of AT use throughout their lives. One of the first steps to better understanding the adoption and design implications of permanent, visible forms of AT is investigating the perceptions that specific populations have that may influence the aforementioned outcomes.

This study focuses on individuals who are single-sided deaf (SSD) and their perception of CIs, which are surgically implanted devices designed to help those who are sensorineurally deaf perceive varying levels of sounds [73, 97] in social interactions with hearing people. SSD is a type of disabled hearing loss that refers to severe-to-profound hearing loss (≥ 70 decibels in hearing level; dB HL) in one ear while having nearly normal hearing in the other ear (-10 to 15 dB HL) [2]. Approximately 466 million people worldwide have some degree of disabled hearing loss, and it is estimated that by 2050 over 900 million people worldwide will have disabled hearing loss [102]. Given that the amount of people with disabled hearing loss is projected to nearly double over the next 30 years, understanding AT adoption and use among members of this community is crucial, as it may help them make more informed decisions about their own AT adoption, as well as inform, and perhaps improve, the design and aesthetics of AT created for hearing loss in the future.

The SSD community, particularly those who have CIs, also provides a useful case for understanding AT adoption and use. First, CIs can only be removed surgically. Due to the risk, cost, and invasiveness of the procedure, CI abandonment is highly improbable. By focusing on individuals' perceptions of a permanent form of AT, this study inverts many of the assumptions of previous AT studies, which focus on preventing AT abandonment and increasing AT adoption and use. Second, SSD is an invisible disability that is rendered visible by CIs. Thus, members of the SSD community are able to directly assess and compare the differences between AT (visible) and non-AT (invisible) interactions with others. Third, because many cases of SSD occur later in life, members of the SSD community who obtain CIs are more readily able to recognize and articulate how permanent AT adoption and use contributes to differences in their interactions. Fourth, examining the SSD community specifically contributes to existing research by focusing on a specific, niche subgroup within the hard-of-hearing (HoH) spectrum. Much of the extant literature on AT adoption and use among individuals with hearing loss tends to view the HoH community as a

monolithic collective. However, there are layers of intragroup dynamics and several subgroups within the HoH community (e.g., d/Deaf, SSD, unilateral hearing loss, etc.), each of which has their own perceptions and orientations toward a variety of factors, including disability identification, social memberships and affiliations, and ATs (like CIs), among others, depending on the nature of their condition, sociocultural norms, and communication needs [52]. Thus, by focusing on a specific subgroup within the HoH community, this study contributes to existing literature by advancing our understanding of a specific subgroup on the HoH spectrum. This population is especially important to consider, as it is understudied relative to other subgroups (e.g., d/Deaf) within the HoH community [17]. Given these factors, the SSD community is a particularly unique and beneficial population to consider, as their experience with CIs provides an empirical case for understanding AT in mixed interactions or interactions that involve “stigmatized and non-stigmatized interactants” [43]. Furthermore, using this group as a case study may provide further insight into the intragroup dynamics among individuals with SSD and the HoH community more broadly, and how AT is perceived and thus, adopted, used, and managed differently within and among members of these groups.

The present study builds upon existing AT and disability literature by examining perceptions of CIs among the SSD community, and how these perceptions relate to levels of perceived self-stigma and socioemotional experiences, namely social interaction anxiety. This study also has broader implications for stigma and AT literature, as it empirically assesses several key concerns of AT adoption (i.e., the relationship between visibility, stigma, and anxiety). Furthermore, by isolating certain factors associated with AT adoption and use, this study helps illuminate whether certain fears about AT in mixed interactions are borne out.

2 RELATED WORK

Our study bridges perspectives from assistive technology, disability, identity, and stigma literature. This section reviews the following areas of prior work: (2.1) the stigmatization of invisible disabilities, such as SSD, (2.2) the stigmatization of assistive technologies, particularly those that are designed for hearing loss (e.g., hearing aids) and (2.3) how the perceived visibility of assistive technologies may contribute to the experience of stigma and social interaction anxiety among members of the SSD community who have CIs.

2.1 Invisible Disabilities and Stigma

Invisible disabilities are disabilities that are hidden or not immediately apparent by an observer except under unusual circumstances or when the individual with the disability discloses about their condition [62]. People with invisible disabilities, like those with SSD, often anticipate and experience stigmatization, particularly in mixed interactions [62]. Stigmatized identities refer to an “undesired differentness from what we anticipated,” where “we” refers to the general society or people who are considered “normal” [39]. Since revealing one’s stigmatized identity can result in several negative outcomes, such as devalued social identities, prejudice, stereotyping, discrimination, and neglect [21, 24, 67], many individuals with invisible disabilities choose to conceal their disability by engaging in passing techniques. Passing refers to the process whereby an individual that perceives themselves as having stigmatized characteristics “manages undisclosed discrediting information about the self” [39] in order to “achieve an identity as a member of the dominant/superordinate group” [15].

A majority of the SSD community acquires disabled hearing loss in one ear later in life, rather than being born with SSD. Therefore, most individuals with SSD must adapt to their disability after a period of time as a hearing person. This means that in addition to the functional changes brought on by acquiring SSD, individuals with SSD must also contend with changes in their personal and social identities. In their social interactions, individuals with SSD are aware of what it is like to interact with others both before and after they acquire disabled hearing loss in one ear, and thus, are more readily able to detect differences in how hearing and disabled hearing individuals are treated in social interactions. Shinohara and Wobbrock [91] found that people who acquire disability later in life or have a lack of experience with AT and disability worry more about the stigma and aesthetics associated with AT adoption and use. In this way, SSD individuals with CIs are a unique population to consider, as their experiences with AT adoption could illuminate the extent to which the perceived visibility of AT affects perceptions of stigma.

2.2 Assistive Technologies Designed for Hearing Loss and Stigma

2.2.1 The Hearing Aid Effect. The extent to which an identity can be stigmatized often depends on the visibility of its “differentness” [47]. AT can make a previously invisible disability visible, as is the case with individuals who are SSD and have CIs. The “hearing aid effect” refers to the negative evaluation of intelligence, achievement, personality, and appearance of people who use AT to aid their hearing [10]. This effect shows there is a stigma associated with hearing impairment, especially when an individual with disabled hearing loss uses AT. Existing research also shows that the hearing aid effect and stigma associated with ATs is applied to anyone with disabled hearing and AT, regardless of age or culture [10, 28, 71].

The hearing aid effect is one of the primary reasons why individuals with hearing AT would abandon their device [55, 84]. Individuals with disabled hearing loss have sociocultural concerns regarding AT adoption and use, such as fear that the devices will attract unwanted attention and that they will be treated differently as a result [27, 76]. AT users can feel even more stigmatized due to the aesthetics of their devices and general lack of acceptance of AT [91]. Faucett and colleagues [32] emphasize that the stigma of AT can negatively impact one’s lifestyle because individuals using AT could feel further exposed or fear facing social consequences like being treated differently. Thus, using AT could result in high levels of perceived stigma, which may then lead to lower self-esteem, social anxiety, overall health [59], and even sustained illness [60].

2.2.2 Perceived Visibility of Assistive Technologies and Stigma. Just as different disabilities have varying levels of visibility, AT visibility also falls on a spectrum. CIs have some variance in size and shape, but more than the objective size of CIs, users’ perception of the visibility of their CI can vary greatly. Self-perceived visibility of AT depends on a multitude of factors, including but not limited to social situations, usage conditions, discourse around the AT, levels of stigmatization, personal attributes, and the degree to which the AT is perceived as being part of their identity [5, 16, 35, 87]. Some scholars note that the self-perceived visibility of AT may be more important to self-image and the personal meaning of AT and one’s disability than the objective size of one’s AT [31, 91]. The personal meaning individuals assign to objects in their everyday life can define identity and affect how they present themselves in social situations [23, 38].

Perceived stigma surrounding AT is also entrenched in social reality. Shinohara and Wobbrock [91] state, “you are who you are perceived to be and that perception can be influenced by what you use.” AT often serves as a signal that the person using it is not an “ordinary person and that one needs to behave differently around this person” [14], which can then influence one’s disability self-

identification [11, 12]. Therefore, there is an emphasis on the force of social expectation and influence on perception as it relates to disability, AT use, visibility, and identity. One study found that observers of individuals with disabilities in public and social spaces evaluated those with visible AT as “others.” The way they explained “otheredness” manifested in different ways, but included those without disabilities pitying or feeling bad for those with disabilities as well as expressing uncertainty about how to interact with them [91]. These findings confirm the stigma of visible AT that individuals with disabilities perceive to be true. Due to the public stigma and perceived self-stigma of AT, a drive for sociotechnical design of AT that minimizes stigma, whether it be by making AT smaller and more “invisible” [42] or more customizable and aesthetically fashionable [82], has emerged. While these designs can be helpful, they may not necessarily address self-perceived stigma or the lived experience of stigma. To better understand this, it is important to examine self-stigma among those with permanent AT, such as members of the SSD community with CIs.

2.3 Perceived Visibility of Assistive Technology

A social identity approach, specifically self-categorization theory (SCT) [91] is a useful lens to consider how and why individuals with disabilities perceive and experience stigma, particularly when they use AT. SCT [96] posits that people self-categorize themselves as a cognitive means of organizing their social reality and that while there are many social categories, only certain ones provide meaning and structure and are deemed salient [45]. According to SCT, people with disabilities may categorize themselves as part of a group with that same disability and its associated meanings and attributes, which will then shape their psychological experiences [94, 96]. Individuals who categorize themselves as a “disability” group member are more apt to be influenced by their group and internalize behaviors and attributes that constitute a prototypical or stereotypical group member [86, 95]. In this way, one’s social identity can lead to emotional and evaluative consequences of group membership, such as developing certain attitudes, perceptions, emotions, and behaviors that line up within a given context.

2.3.1 Perceived Visibility and Self-Stigma. In the context of SSD, particularly for those who have CIs, AT can serve as a marker of hearing loss and thus, reinforce stereotypes associated with hearing loss and AT use. Furthermore, the visibility of AT designed for hearing loss can make one’s identity as SSD even more pronounced. Given what we know about the perception of visibility of AT and its effects, stigma in the context of disabilities and AT, and SCT, we predict that SSD individuals with CIs who perceive their AT to be highly visible will experience greater levels of self-stigma:

H1: Perceived visibility of one’s AT is positively associated with self-stigma in mixed interactions.

2.3.2 Perceived Visibility & Social Interaction Anxiety. In mixed interactions, the expectation and fear of rejection and ostracism may increase stress and anxiety, particularly social anxiety. Social anxiety refers to experiencing discomfort in certain social and interpersonal situations [58]. It can result from the anticipation, prospect, or presence of personal evaluation in imagined or real social situations [90]. According to the cognitive self-evaluation model of social anxiety [44, 85], social anxiety does not stem from objectivity, but from individuals’ perception of personal inadequacies. Therefore, perceived, anticipated, or imagined reactions can all equally contribute to social anxiety in mixed interactions. AT, such as CIs, can serve as a social symbol or artefact or sign, that denotes disability [1]. AT as a social symbol may affect social interactions and anticipations of negative evaluation from interactants, particularly because visible AT can serve as a signal of membership to

a stigmatized (i.e., disabled, hard-of-hearing) group [90, 92]. Independent of the extent to which one's AT is visible in social interactions, perceptions and perceptual barriers can create real barriers in mixed interactions for those who are disabled [35]. Hence, we predict that higher perceived visibility of one's AT is positively associated with social anxiety in mixed interactions:

H2: Perceived visibility of one's AT is positively associated with social interaction anxiety in mixed interactions.

3 METHODS

3.1 Procedure

This study is part of a larger project that employed an online survey. Participants were recruited from SSD social support groups on Facebook. Moderator approval and confidentiality forms were required for certain groups, after which a recruitment letter on the group's main Facebook page with a link to the online survey hosted by Qualtrics was posted. In support of research on the SSD community, the moderators or administrators of these groups often pinned our post to the top of the discussion wall and/or encouraged group members to complete the survey. This was especially useful as the SSD community is a hard-to-reach population.

The study was approved by the Institutional Review Board (IRB) of the authors' affiliated University. Participants gave informed consent on the first page of the survey before completing the questionnaire. After taking the survey, participants were given a debriefing form. They were also thanked for their time. The survey instrument was constructed using a combination of established and validated scales as well as original items, which we discuss below. Prior to data collection, the survey was pilot tested with a subsample of 17 people in order to clarify question wording. Data collection for this project took place from Spring 2019 to Fall 2019.

3.2 Participants

Inclusion criteria limited the sample to individuals 18 years and older who were clinically diagnosed with SSD, acquired SSD at some point in their life (meaning they were not born with SSD), and had a CI. A total of 332 participants completed our survey. Participants were from all over the world and reported residing in different states across the U.S. including Rhode Island, Georgia, Arkansas, Illinois, New Mexico, Nebraska, and California, among many others, as well as various countries around the world including, but not limited to the United Kingdom, Canada, Australia, Kenya, Bangladesh, the Philippines, Brazil, and New Zealand. Of the participants, 59.6% identified as female, 37.7% identified as male, and 2.7% identified as something else. Participants ranged from 21- to 79-years-old ($M=50.23$, $SD=12.94$). Most of our participants were unable to identify the cause of their SSD, but for those who were, causes of SSD included acoustic neuroma, bacterial meningitis, and Ramsay-Hunt Syndrome. The length of time participants reported living with SSD ranged from just under 1 year to 65 years ($M=16.67$, $SD=13.64$).

3.3 Measures

3.3.1 Visibility of Cochlear Implant as Assistive Technology. On a scale from 1 (*not visible at all*) to 10 (*very visible*), participants were asked how visible they thought their CI was to other people in social interactions.

3.3.2 *Self-Stigma.* The Stigma Scale [48] was used to measure participant perceptions of self-stigma. This study used the shortened version of the original stigma scale [48], which resulted in a 28-item scale. Items were adapted to refer to SSD and included statements like, “Other people have made me feel ashamed of myself because of my SSD,” “I have been discriminated against because of my SSD,” “Sometimes I feel that I am being talked down to because of my SSD,” “People have avoided me because of my SSD,” “Very often I feel alone because of my SSD,” “People’s reactions to my SSD make me keep myself to myself,” “I feel embarrassed because of my SSD,” “I avoid telling people about my SSD,” and “I feel the need to hide my SSD from my friends.” Participants indicated their level of agreement with each statement using a five-point Likert scale (1=*strongly disagree*, 5=*strongly agree*).

3.3.3 *Social Interaction Anxiety.* The level of anxiety participants experience in social interaction was measured using the 20-item Social Interaction Anxiety Scale [63]. On a scale of 1 (*not at all characteristic or true of me*) to 5 (*extremely characteristic or true of me*), participants were asked to indicate the degree to which they felt each statement was characteristic of or true for them. Items included, “I feel tense if I am alone with just one other person,” “I have difficulty talking with other people,” “I worry about expressing myself in case I appear awkward,” “When mixing socially, I am uncomfortable,” and “I tense up if I meet an acquaintance in the street,” among others.

4 RESULTS

A multivariate linear regression analysis was conducted to test H1 and H2. Before analyzing the research hypotheses, we examined relationships between our variables with bivariate correlations as demonstrated in Table 1. All multivariate analyses were 2-sided with a significance threshold of $p < .05$. To remain consistent with other studies as well as for theoretical reasons, our analysis treated age [63], years SSD [11], and gender [100] as covariates.

Table 1. Bivariate Correlations Among Research Variables (N=332)

Variable	1	2	3	4	5
1. Age	—				
2. Years SSD	.22**	—			
3. Visibility	-.04	-.06	—		
4. Perceived Stigma (PSOSH)	-.03	-.01	.36**	—	
5. Social Anxiety (SAIC)	-.02	.01	.49**	.29**	—

* $p < .05$ ** $p < .01$ *** $p < .001$.

To test H1 and H2, we performed a multivariate linear regression analysis, treating social interaction anxiety and self-stigma as outcome variables. H1 predicted that perceived visibility of one’s AT is positively associated with self-stigma in mixed interactions. Consistent with H1, the perceived visibility of AT and self-stigma was significant, $B = .13$, $SE = .02$, $p < .001$, showing that the more individuals with SSD perceived their CI to be visible to others in mixed interactions, the more stigma they felt. H2 predicted that perceived visibility of one’s AT is positively associated with

social interaction anxiety in mixed interactions. Consistent with H2, the results indicated a positive association between perceived visibility of AT and social anxiety in mixed interactions, $B = .14$, $SE = .01$, $p < .001$, revealing that the more individuals with SSD perceived their CI to be visible to others in mixed interactions, the more social interaction anxiety they felt.

5 DISCUSSION

5.1 Overview of the Association between Perceived Visibility of Assistive Technology, Self-Stigma, and Social Interaction Anxiety

This study examined members of the SSD community who have CIs and how their perception of visibility of AT affects self-stigma and social anxiety in mixed interactions. Our results show that the more individuals perceive their AT to be visible, the more self-stigma and social anxiety they experience in mixed interactions. These findings contribute to our theoretical understanding of AT by emphasizing the social meaning of these technologies as perceived by the user in mixed interactions, which is critical to understanding the viability of AT for people with disabilities, particularly those that are permanent, voluntary, and visible.

Several studies address AT abandonment [33, 78, 93], including how aesthetics of AT relate to stigma, how AT users are perceived in society, and how perceptions of AT users affect device abandonment [75]. In the context of disabled hearing loss, many people refuse to use AT due to cost, discomfort, sense of foreignness, stigmatization [51], social rejection [98], and shame. These studies have led to a focus on identifying the factors that lead to AT non-use and/or abandonment as a means of developing ways to increase AT use and adoption [99]. The case and context of our study, however, provides new dimensions for understanding voluntary, permanent, visible AT and its association with self-stigma and social anxiety. Specifically, our results show that the more users perceive their AT as being visible, the more self-stigma and social interaction anxiety they experience as a result. This is especially important to consider in the context of CIs and other forms of visible, permanent AT, as adopters of these devices are not able to easily abandon or discontinue using them, and thus, must continually contend with and manage the consequences of AT adoption, including but not limited to increased self-stigma and social interaction anxiety.

5.2 Moving Beyond Visible/Invisible towards Visibility as a Spectrum

The results of this study also contribute to stigma and disability literature by problematizing the assumption that people with invisible disabilities have more advantageous attributes, characteristics and experiences than those with visible disabilities [47]. Existing studies show that people with invisible disabilities often engage in “passing” by deliberately concealing their disability to avoid negative attitudes, stigma, and discrimination [18, 19, 74, 83]. The ability to “pass” is considered an advantage that individuals with invisible disabilities (such as SSD) have and that those with visible disabilities do not. However, our study demonstrates that invisible disabilities can become visible through AT and that adopters of these devices experience more self-stigma and social anxiety simply due to the *perception* of their AT as being visible. In this way, this study not only confirms that objective visibility (e.g., size, color, placement, etc.) is critical to the experience of stigma, but that perceived visibility is central to this process as well. Even when individuals adopt AT that is objectively less visible (e.g., small in size, can be placed or worn underneath hair or clothes, etc.), they may still experience significant levels of self-stigma and social anxiety simply due to the perception that their AT and thus, disability, is visible to others. Given these findings, future research should avoid objective categorizations of both disabilities and AT as either invisible or

visible. Instead, our results suggest that both disability and AT should be viewed on a spectrum, where both objective and perceived visibility affect levels of stigma and social interaction anxiety. Additionally, future studies on the visibility of disability and stigma should consider AT as a moderating, even accentuating factor, given the role they play in the socioemotional experiences of individuals with disabilities.

By highlighting how the perceived visibility of AT is associated with higher levels of stigma and social interaction anxiety, this study also emphasizes how addressing the functional components of disability (hearing loss) influences the perception of AT (CIs) and consequently perception of the self (self-stigma). This confirms the importance of perceptions of AT and visibility and how they affect self-image as well as the personal meaning attributed to AT and one's disability [91]. CIs are a permanent, voluntary, visible form of AT that are designed to enhance the physical functionality (ability to hear) and even social functionality (ability to interact with others) of individuals with disabled hearing loss in mixed interactions. However, due to the permanent, visible nature of CIs, individuals with SSD that adopt these devices may be permanently "marked" as disabled by others and thus, forced to identify as disabled in social interactions. This demarcation poses a variety of concerns related to stigma and social presentation in mixed interactions, which may inherently limit or even reduce the functional benefits offered by AT. Thus, our study emphasizes how perception of visibility is a critical variable in assessing how people with SSD that have CIs navigate mixed interactions, a variable which may also be applicable to other forms of invisible disabilities that are rendered permanently visible through AT adoption and use (e.g., prosthetics).

5.3 The Value of a Social Identity Approach in Assistive Technology Research

The results of this study also illuminate the utility of a social identity approach, specifically SCT. Not only does SCT accurately predict and explain how perceptions of visibility of AT shape the meaning attributed to these devices and self-stigma, it also provides a dynamic view of how social categories provide meaning and salience in different contexts. Specifically, this study demonstrates how the perceived visibility of AT, rather than its objective features, impacts disability self-identification and psychosocial and communicative factors in mixed interactions. This study examines a specific population that uses the same technology, but experiences it differently based not on the objective features and characteristics of the technology itself and its affordances, but rather, on the perception of the technology. This is consistent with other studies that found perception of limited capabilities associated with disability was among the main reasons that kept people with disabilities interacting with others [54].

Stigma is associated with AT usage, particularly for individuals with disabilities acquired later in life [13, 104]. Additionally, SCT may be used to consider how the intersectionality of stigmatized identities (e.g., race/ethnicity, having more than one disability, having more than one AT, etc.) shapes how individuals who perceive their AT to be highly visible engage in self-categorization. Individuals have membership to various groups and thus, identify with several group categories. Disability is but one of many stigmatized identities and the intersectionality of this identity with other stigmatized identities could result in an amplification of stigma [4]. For example, the intersection of disability and gender [34, 37, 68], race [8, 6, 64, 70], income [8], and age [25] among others, influence disability self-identification such that there is a likelihood of double oppression or exposure to various types of stigma. Using a social identity approach, future studies could evaluate the role that disability and AT play with the intersection of other identities and its association with stigma.

The results from this study also illuminate the importance of examining how the adoption and use of technology may affect members of the same groups and communities differently. In the context of SSD, scholars tend to study individuals with SSD as a collective. Given the variety of conditions and sub-communities within the hard-of-hearing spectrum (e.g., deaf, Deaf, SSD, unilateral hearing loss, etc.) as well as the nuances in their social, cultural, and communicative norms, these differences are important to account for and understand, as the SSD community is highly complex and multi-faceted, rather than monolithic. The nuances among this community can provide insights about how members of the same group use and perceive the same technologies differently. This study sheds light on how each group and subgroup may have their own orientation towards technologies like CIs depending on the nature of their condition and communication needs.

Our study also quantitatively confirms the cognitive self-evaluation model of social anxiety, where perceived, anticipated, and imagined visibility of AT affects social anxiety, and highlights the interpersonal communicative aspects of social anxiety. Furthermore, it points to the possibility that there may be a feedback loop in which individuals feel heightened perceptions of visibility of AT and social interaction anxiety. Beyond the direct associations measured in our study (H1 and H2), the bivariate correlations among our research variables indicate that there is a positive association between self-stigma and social interaction anxiety, which may explain the possible socially recursive feedback loop between visibility of AT, stigma, and anxiety. Future studies should investigate the causal relationships of this socially recursive process to better understand the directionality and effect of these variables.

5.4 The Implications of Perceived Visibility on Design

This study also has practical implications, particularly regarding AT design. Although there are approaches to AT design aimed at increasing the personalization and customization and thus, variance in visibility of AT [9], utilizing matching technologies frameworks [89], and identifying inclusive accommodation factors [7], there is a general lack of sociotechnical pairing between people with disabilities and the AT they use. Furthermore, there is oftentimes a momentum to “normalize” interactions for people with disabilities by using technological interventions, but this strategy based on compensation could further highlight boundaries between people who are disabled and those who are not [41, 69]. As a result, many AT users do not feel positively about their AT or AT use, which may result in several negative outcomes. For this reason, our study suggests there is a need to consider the sociotechnical aspects of AT during the development of these devices, rather than post-hoc. Ultimately, this calls for participatory approaches to AT design that emphasize designing *with*, rather than *for* users [81, 88]. Tailoring to the needs of the user of AT through user centric techniques, AT designers may be able to better understand and thus, account for the sociocultural and psychological practices and expectations associated with people who have disabilities, and ultimately, co-construct and develop AT that better serves their needs and improves their quality of life [3]. In the context of SSD and CIs, for instance, a participatory design approach could be used to develop hearing loss AT that is perceived by users as being less visible, which our results suggest may decrease self-stigma and/or social interaction anxiety. This approach to design is particularly important to consider in the context of CIs and other forms of visible, permanent AT, which are difficult to remove, and thus, must be continually used and managed by users after adoption. This sociotechnical approach could also highlight questions surrounding emerging social technologies, equality, diversity, and design, as disabilities themselves could reveal and illuminate new information about technologies and how there may be new perspectives and framings of communication and technology [40]. Future studies could also evaluate the rationale behind design

features of ATs like CIs, as representations of disabilities like deafness can influence technological changes while limiting certain functions and holding institutional biases that perpetuate marginalization and constrain communication [30].

5.5 Limitations

As with all studies, our study has some limitations. Our sample only consisted of people who acquired SSD later in life and thus, did not consider the perspectives of individuals who were born with SSD, which may affect the perceived visibility of AT and subsequently, the level of self-stigma and social anxiety experienced. Future work should further investigate the relationship between perceived visibility of AT and stigma and social anxiety by comparing individuals with congenital and acquired SSD to see if time, self-identification with disability, and relationship with AT affects or moderates levels of self-stigma and social anxiety. Additionally, our study only considered one type of AT designed for hearing loss: CIs. Individuals who have disabled hearing loss can adopt various technological hearing interventions, some of which are more easily concealed and/or removed than others. Future studies should evaluate whether the AT itself and its degree of permanence and objective visibility (e.g., size, color, shape, etc.) changes perception of visibility across AT designed for hearing loss. Since this study used an online survey, we were also unable to explore why, how, and which scenarios our participants imagined, anticipated, and/or perceived in their mixed interactions. Qualitative studies, specifically interviews, could help us better understand the processes individuals with AT go through when making sense of their AT in mixed interactions as well as their experience of self-stigma and social anxiety. Furthermore, since this study was part of a larger project and due to time constraints, we were unable to ask our participants more objective questions about their disability and technological efficacy (e.g., actual hearing ability, efficacy of their AT, etc.), which could also influence their level of self-stigma and social interaction anxiety. It would be interesting for future studies to replicate or extend this study by including and combining more objective measures with the perception-driven measures. Lastly, sampling from Facebook groups is a limitation, as the members of these communities are not necessarily representative of the larger SSD population. There may be important differences between individuals who are members of these communities and those who are not, including but not limited to variances in their level of self-stigma and social anxiety, support seeking behaviors, perceptions of AT, and experiences with disability.

6 CONCLUSIONS

This study examined perceptions of visibility of AT and how these perceptions contribute to self-stigma and social anxiety in mixed interactions. Investigating the role of perceptions is important as adoption and usability of AT is dependent on these perceptions [79]. Results indicate that the more users perceive their AT as being visible, the more self-stigma and social anxiety they experience. These results not only highlight the centrality of the *perception* of ATs to the experience of stigma, but also problematize the idea that visibility is dichotomous (i.e., visible vs. invisible) and objective. Rather, our study suggests that visibility exists on a spectrum and that ATs may moderate the degree to which an individual *perceives* their disability as being visible to others, a factor which our study suggests is equally important to consider as the objective visibility, characteristics, features, affordances, and nature of the AT itself. Despite the functional benefits offered by CIs, such as the ability to hear and interact more effectively in mixed interactions, our study suggests that, in some cases, the stigma associated with and the perception of these devices can lead to psychological

and social outcomes, such as increased levels of self-stigma and social anxiety in mixed interactions, that may complicate these benefits. Furthermore, much of the CSCW community has found that different groups will adopt the same technologies for different purposes. This study also highlights the need to and utility of examining niche stigmatized communities that use the same assistive technologies, as they may provide further insight into how the use, adoption, and experience of ATs can vary due to the characteristics of certain conditions and/or disabilities, as well as the variety of meanings ascribed to these technologies among different members of the same group. These findings are also important to consider from a design perspective. Specifically, our study illuminates the need for participatory design that considers the perspectives and experiences of AT users and attempts to account for not only the functional, but the sociocultural and psychosocial aspects of AT as well. In doing so, we may be able to develop AT that goes beyond functionality and addresses the sociocultural consequences posed by AT adoption and use, which may help enhance the well-being and quality of life for individuals living with disabilities.

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